

Witness the Flow in Tennis

Summary

Tennis, as a sport filled with uncertainty, sees a crucial impact from a player's individual "momentum" on the course of a match. Our study delves into the mechanisms and effects of the "momentum" factor in tennis, establishing a scientific **model for capturing momentum fluctuations**. This model aids players in understanding the influence of momentum on matches, assisting them in formulating effective strategies. Simultaneously, it contributes to coaches in creating more scientifically grounded training plans, genuinely elevating players' competitive levels.

In addressing the first question, we introduced the concept of momentum based on actual match scenarios, empirically analyzing its impact on the course of a match and affirming the feasibility of our study. Subsequently, we established indicators for momentum-related factors, conducted regression analysis, and verified the results. We then extracted relevant indicators, built a model for evaluating players' competitive levels, and utilized **factor analysis** to reduce dimensionality. Building upon this analysis, we constructed a **weighted LSTM neural network state prediction model** based on momentum factors, achieving a model fitting goodness of **95.02%** in testing and deeming the model highly accurate.

For the second question, we initially conducted a correlation analysis of the indicators, concluding that momentum independently affects a player's scoring situation. We proceeded to analyze how momentum influences the course of matches, challenging coaches' perspectives. Employing **Monte Carlo methods**, we randomly generated momentum data for multiple simulations, revealing a deviation degree of **93.21%**. Contrasting this with our established momentum impact model, we refuted the coach's viewpoint on a data-driven basis.

In addressing the third question, we analyzed indicators influencing fluctuations, classifying them for discussion. Through methods such as **multiple linear regression**, **entropy weight**, and **TOPSIS**, we established five indicators for a comprehensive model of influencing factors. Introducing the concept of state transition distribution, we built a **fluctuation prediction model based on Markov chains and multiple nonlinear regression**. Visualizing and analyzing the results confirmed the model's rationality, concluding with proposed strategies for athletes in different situations.

For the fourth question, we initially validated the model by predicting outcomes in the top 16 matches, achieving an accuracy rate of over **85%**. Considering other potential influencing factors, we extended the model's application to table tennis matches. After adaptively modifying the model and testing it on new data, we found the model remained relatively accurate. Finally, we expanded the model to women's matches, championships, and different court surfaces, demonstrating its robust generalization and applicability.

Keywords: Multiple Linear Regression, LSTM Neural Network, Monte Carlo Method, Markov Chain.

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1 Introduction

1.1 Problem Background

Tennis is a sport characterized by both high competitiveness and entertainment value. It not only tests the players' athletic abilities but also involves a comprehensive competition of tactical strategies and mental resilience. The intense confrontational nature of this sport makes the factor of "momentum" crucial and impossible to ignore.

In the 2023 Wimbledon Gentlemen's final, Carlos Alcaraz defeated Novak Djokovic. Despite being an experienced player who was expected to win easily, the seasoned veteran ended up losing the match after a fierce competition. The young player, relying on his incredible momentum, emerged victorious in the face of strong opposition. Due to several crucial fluctuations during the match that ultimately altered its course, we aim to establish models and analyze these phenomena based on Wimbledon_featured_matches .

This research aims to explore the specific impact of momentum on match outcomes. By doing so, it seeks to assist players in better understanding and leveraging the favorable effects of momentum, enabling them to adjust their tactics promptly during matches. Through a rational use and alteration of momentum, players can achieve better results in their performances.

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- **Problem 1:**After considering the impact of serving factors, analyze players' performance in matches and establish a scoring flow capture model. The model should be capable of assessing players' performance levels and capturing the flow of scoring during a match. Next, apply the model to analyze players' status levels and visualize the performance results in one or more matches.
- **Problem 2:**Utilize the established model to evaluate the role of momentum in matches. Analyze the relationship between swings in play and runs of success by one player, and address the queries of tennis coaches.
- **Problem 3:**Using data from at least one match, construct a model to predict score fluctuations in matches. Analyze the factors related to fluctuations, indicating when a match will shift from favoring one player to favoring another. Provide strategic guidance for players facing each other in new matches based on past match data.
- **Problem 4:**Test the fluctuation prediction model in more matches, assess its accuracy, and analyze whether additional data support is needed. Further, generalize the model, analyzing its applicability to other types of matches.
- **Problem 5:**Summarize the model and then write a report and memorandum. Provide recommendations to coaches based on the model's findings regarding the impact of "momentum." Additionally, offer strategic guidance for events that impact the flow of play during a tennis match.

1.3 Our Work

- **Step1:**In addressing Problem 1, we need to establish a model capturing the flow of scores during a match. We will select and extract relevant data for analysis. By identifying key factors based on match characteristics, we will establish evaluation metrics for players' athletic levels. Subsequently, we will analyze score flow characteristics based on these metrics and build a scoring flow capture model.
- **Step2:**For Problem 2, which requires assessing the role of "momentum" in matches, we will quantitatively analyze the relationship between momentum and players' scoring abilities. We'll extract match data to visualize the real-time relationship between momentum and scores. Additionally, we'll compare predictive abilities under random conditions with those under actual momentum influence to demonstrate the rationality of our momentum analysis. This will serve to counter the coach's viewpoint.
- **Step3:**In addressing Problem 3, where we need to establish a model to dynamically determine shifts in advantage, we will extract factors influencing players' scoring abilities. Using Markov chains, we will calculate the distribution of players' inherent advantages. For other factors, a multivariate nonlinear regression will be performed. Subsequently, we'll establish player state fluctuation models and advantage state fluctuation models to determine shifts in advantage. Strategies for on-court responses under different circumstances will also be proposed.
- **Step4:**Problem 4 requires analyzing the model's accuracy in multiple matches and generalizing it to different types of matches. We'll initially test and evaluate the established model using data from Wimbledon_featured_matches. Following that, we'll test the model in matches of various types, making improvements based on the characteristics and differences of these matches.

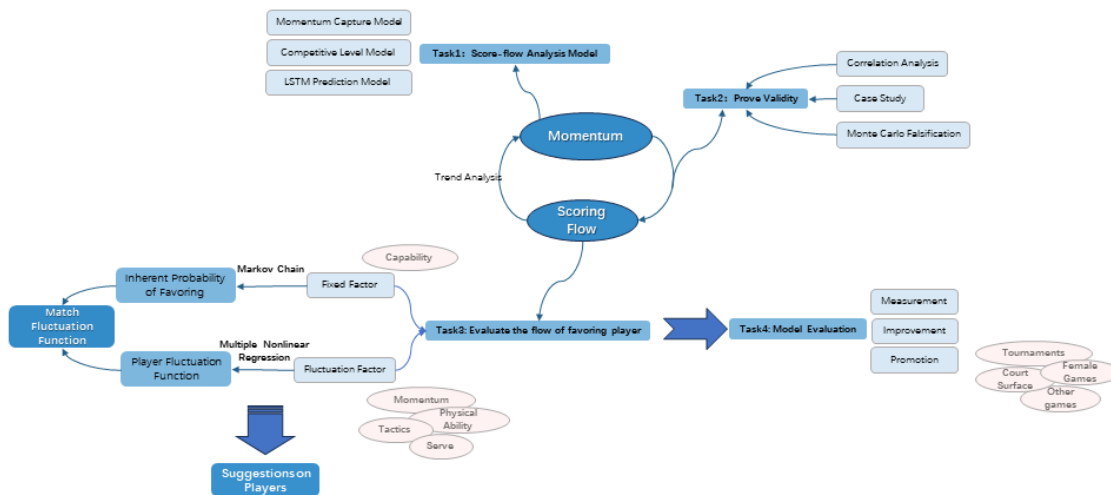


Figure 1: Our Work

2 Preparation for modeling

2.1 Assumptions

1. Assuming that other influencing factors in the environment of both players in the same match (such as lighting, court conditions, noise, etc.) are the same, and the players are not affected by injuries during the game, they have both completed the match in their optimal condition.
2. Assuming that the generation of momentum is influenced by crucial victories and specific match outcomes (such as aces), and the impact of the same factor remains constant at different times.
3. A large amount of data indicates that momentum's effect can only be short-term. Therefore, we assume that momentum in a match is a relative change and cannot grow indefinitely.
4. As the impact of "momentum" in competitive sports is consistently positive, we assume that the effect of "momentum" can be generalized to other matches.
5. Assuming the growth and decline of momentum follow a continuous process.

2.2 Notations

Table 1: Notations

Symbol	Definition
M	Player Momentum Tracking Model
Ct	Memory Cells
C	Player Performance Rating
T	Tactical Richness Evaluation Model
Ps	Physical Fitness Assessment Model
M	Player Momentum Assessment and Tracking Model
S	Serve

2.3 Data Cleaning

Step1:Missing value imputation

1. For the variable "speed_mph," representing serve speed in miles per hour, data for matches 1310 and 1311 are excluded from analysis due to excessive missing speed information after excluding second serve faults. The remaining missing data are imputed with the mean value.
2. Data for "rally_count" in matches 1310 and 1311 are missing, so these two matches are excluded from analysis.
3. Considering significant information loss in "return_depth," missing values are defined as having no apparent tactical changes.
4. For "serve_depth" and "serve_direction," where missing data are minimal and dispersed, this study defines them as having no apparent tactical changes.

5. "distance_run" and "rally_count" have relatively few missing values, and the mean values are used for imputation.

Step2:Data Transformation

Due to the transient nature of momentum, the impacts of these behaviors are categorized into two types. A damping model is established to simulate the effects of these behaviors on players.

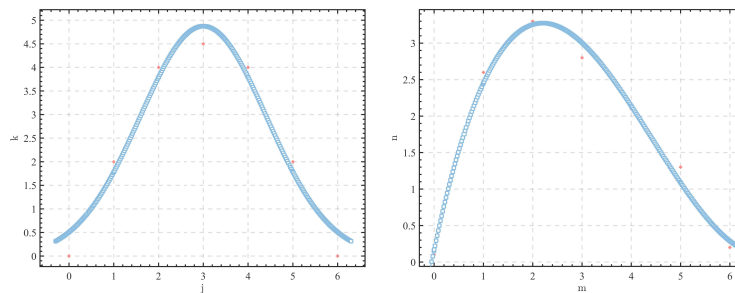


Figure 2: damping model

1. Application of Damping Model 1:

In tennis, players gain a significant boost in momentum after hitting an ACE, directly scoring a point. For events such as ACE (positive impact) and double_fault (negative impact), we applied Damping Model 1 to simulate the player's momentum changes.

2. Application of Damping Model 2:

Certain common player behaviors in tennis, such as unf_err, have a relatively slow impact on the player's momentum. Damping Model 2 is applied to simulate this gradual effect on player momentum.

3. Impact of Continuous Serving and Return Strategies:

The consecutive serving and return strategies of players, involving combinations of speed, direction, and depth, have a substantial influence on the probability of scoring. We measure the tactical changes of players by assessing the variations in these aspects over five consecutive instances.

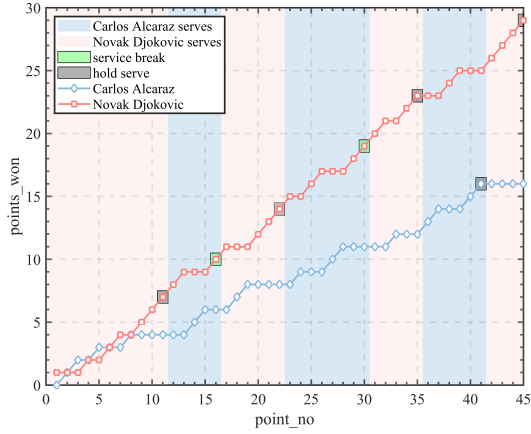
4. Consideration of Short-Term Fitness Factors:

Different players experience varying levels of physical exertion over short periods, and younger players may be less affected by the same sprint distance. We analyze fitness factors by transforming them into average levels over five consecutive points for each player.

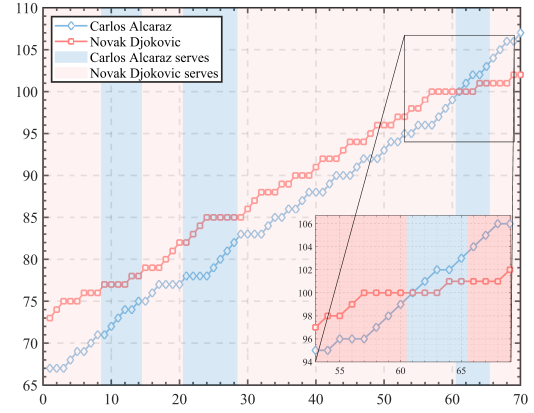
3 Resolution of Problem I

3.1 Analysis of Match Trends

Given the variations in players' playing styles, skill levels, and the unique circumstances of each match, aggregating data from multiple matches inevitably introduces uncertainty into the analysis results. Therefore, we extracted data from the first and third sets of the 2023 Wimbledon match between Alcaraz and Djokovic:



(a) First Set



(b) Third Set

First Set: In the initial set, Djokovic consecutively broke Alcaraz's serve in two games, establishing a formidable momentum that allowed him to easily secure victory in the first set.

Third Set: In the third set, Alcaraz quickly gained points while Djokovic was serving, securing consecutive victories and maintaining a strong momentum. Throughout the subsequent match, he easily emerged victorious, achieving a comeback. Contrasting Djokovic's dominant performance in the first set with Alcaraz's crucial reversal in the third set highlights the significant impact that different momentum levels can have on a player's scoring ability. Our team believes that studying the influence of momentum on players' scoring abilities in a match is crucial.

3.2 Establishment of Scoring Flow Analysis Model

Through the aforementioned analysis, we identified that momentum significantly influences a player's actual scoring ability. Therefore, we have developed a model to quantitatively analyze how momentum affects a player's scoring ability.

3.2.1 Player Momentum Capture Model (M)

Initially, we captured factors influencing the original momentum in the match and conducted an analysis:

Positive Influencing Factors:

$$M_1 = \lambda_1 f(m_1) + \lambda_2 f(m_2) + \cdots + \lambda_5 f(m_5) = \sum_{i=1}^5 \lambda_i f(m_i) \quad (1)$$

Negative Influencing Factors:

$$M_2 = \lambda_6 f(m_6) + \lambda_7 f(m_7) + \lambda_8 f(m_8) = \sum_{i=6}^8 \lambda_i f(m_i) \quad (2)$$

$$M = M_1 - M_2 \quad (3)$$

λ_1	Whether scores in a row	λ_5	Whether he broke serve
λ_2	Whether to serve the winner	λ_6	Whether he is behind
λ_3	Whether it is an untouchable shot	λ_7	Whether there are double faults
λ_4	ACE number	λ_8	Unforced errors count

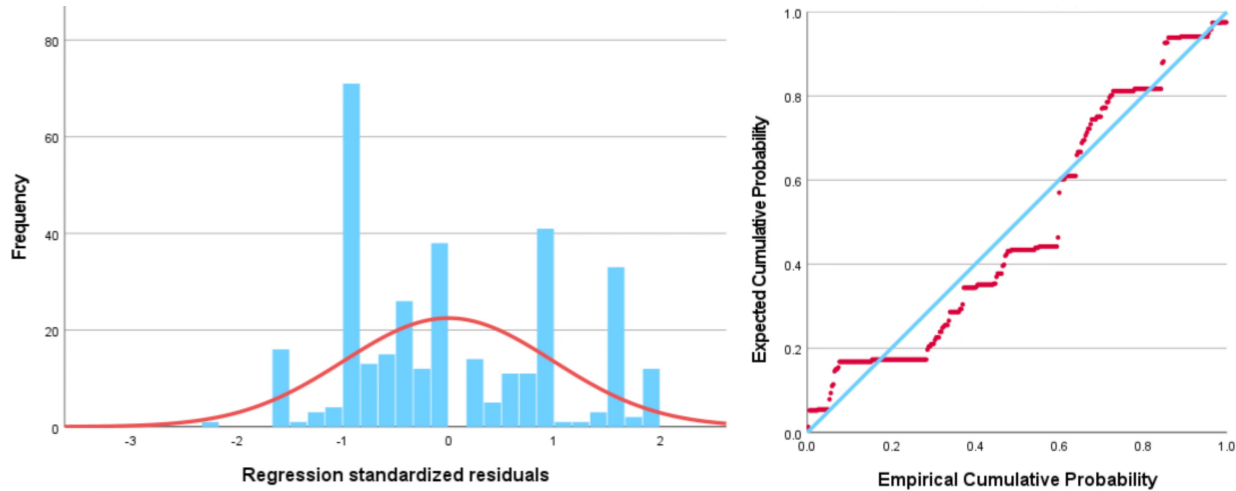


Figure 4: Residual Histogram and Normal P-P Plot of Regression Standardized Residuals

1. The plot aligns with the normal distribution curve, and the standard deviation of residuals is 0.994, indicating that the residuals follow a normal distribution.
2. The Durbin-Watson statistic yields a value of 2.055, suggesting that the samples exhibit independence.
3. The Variance Inflation Factor (VIF) values for all eight factors are below 10, indicating the absence of multicollinearity among the factors.

Based on the above tests, it is evident that multiple linear regression can be applied to the eight parameters utilized in this model and the momentum in the final. The significance level (P-value) of 0.000*** demonstrates statistical significance, confirming the effectiveness of the model.

parameter	coefficient	standard error	Wald	P	Lower bound of 95% confidence interval	Upper 95% confidence interval
intercept	1.168	0.167	48.853	0.000***	0.841	1.496
consecutive scores(Yes_1;No_0)	0.101	0.016	37.457	0.000***	0.069	0.133
Number of ACES(damping)	0.109	0.035	9.67	0.002***	-0.178	-0.04
Break or not(damping)	0.066	0.031	4.526	0.033**	0.005	0.127
Double error or not(damping)	0.029	0.03	0.933	0.334	-0.03	0.089
Number of unforced errors	-0.176	0.025	47.803	0.000***	-0.226	-0.126
Service winner(Yes_1;No_0)	-0.24	0.169	2.007	0.157	-0.572	0.092
Winner or not(Yes_1;No_0)	-0.488	0.049	97.121	0.000***	-0.585	-0.391
Lag behind(Yes_1;No_0)	-0.018	0.033	0.289	0.591	-0.084	0.048

Table 2: Multiple Linear Regression Results

Upon comparing significance levels, it was found that consecutive scoring, the number of ACEs, breaking serve, the number of non-forced errors, and winning shots significantly influence the momentum for both sides. Therefore, the momentum regression analysis model is as follows:

$$M = 1.168 + 0.101m_1 - 0.488m_3 - 0.109m_4 + 0.066m_5 - 0.176m_8 \quad (4)$$

During the analysis of the regression results, an interesting phenomenon was observed. In the final match, the number of ACEs showed a negative correlation with the scores of both players, contrary to the common sense that hitting ACEs should motivate players.

To address this, we extracted and analyzed the relationship between the number of ACEs and scores in the quarterfinals. In matches with a significant difference in strength, the number of ACEs exhibited a weak positive correlation with scoring. However, in closely contested matches, the results aligned with our model. To enhance the accuracy of the model in predicting closely contested matches, no modifications were made.

In the final, Alcaraz hit a total of 9 ACEs, but in the decisive game, hitting an ACE resulted in consecutive points lost. We hypothesize that, at this point, both players were at a high level of momentum, making it easier to produce high-quality serves. However, this behavior carries risks, as high-quality serves may lead to equally high-quality returns from opponents, potentially diminishing the advantage gained from serving. Additionally, opponents may showcase better performance in the subsequent points to thwart the player's momentum. We speculate that hitting ACEs may have a negative impact on the momentum of high-level players, potentially causing consecutive points losses.

3.2.2 Establishment of Player Performance Evaluation Model

Step 1: Data Feature Extraction

Symbol	Description	Symbol	Description
x_1	first-serve successful rate	x_{11}	serve direct rate
x_2	first-serve scoring rate	x_{12}	double-fault rate
x_3	second-serve scoring rate	x_{13}	scoring rate while at the net
x_4	first-serve direct rate	x_{14}	break successfully rate
x_5	serve scoring rate	x_{15}	unforced error rate
x_6	success rate when opponent is in first-serve time	x_{16}	average speed of first-serve
x_7	success rate when opponent is in second-serve time	x_{17}	average speed of second-serve
x_8	scoring rate when opponent is in second-serve time	x_{18}	maximum speed of serve
x_9	catching game winning rate	x_{19}	untouchable shot count
x_{10}	break point count		

Based on the provided table, and for the purpose of facilitating the evaluation of scoring flow data for each player, the data features have been extracted as shown in the table above.

Step 2: Correlation Analysis of Indicators

Initially, a correlation analysis was conducted for all the indicators.

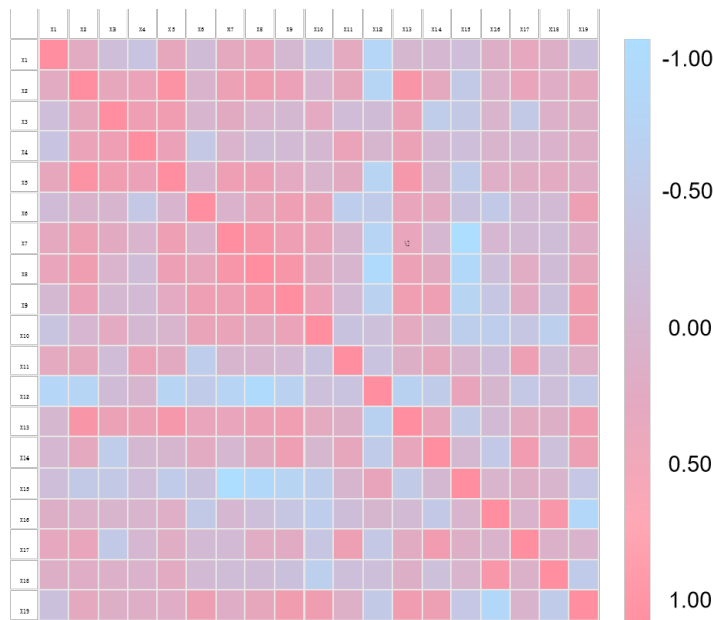


Figure 5: Correlation Analysis Heatmap

We observed a significant level of collinearity among these indicators. Consequently, we performed the following dimensionality reduction on these nineteen indicators.

Step 3: Dimensionality Reduction Based on Factor Analysis Method

We conducted a cluster analysis on the indicator data and subsequently utilized the clustering results as a reference for principal component analysis. The analysis results are as follows:

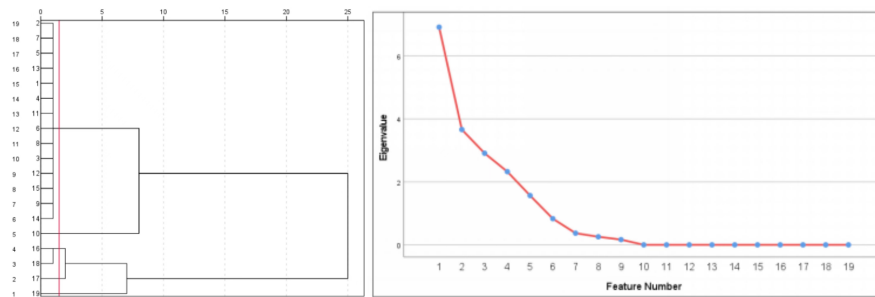


Figure 6: Cluster Dendrogram and Scree Plot of Principal Component Analysis

By observing the cluster dendrogram and the scree plot of principal component analysis, we determined that the clustering is most effective when the cluster step is set to 12, resulting in 5 clusters.

Our analysis results are as follows: The weight of success rate when opponent ,break point count,average speed of first-serve,average speed of second-serve,untouchable shot count are :

20.2233% 33.4686% 12.9507% 17.5372% 15.8201%

Based on the clustering results and principal component analysis, the sum of the contributions of these five indicators to player scoring reached 91.45%, suggesting that these metrics can effectively describe a player's current competitive state, aligning with the actual evaluation methods used in practice. With this, the establishment of the indicator model is complete.

3.2.3 Weighted Selective LSTM Neural Network Score Prediction Model Based on Momentum Factor

The analysis indicates that whether a player is serving, the player's skill level, and the "momentum" factor significantly influence a player's scoring ability. As these factors have a temporary impact on scoring ability, we have improved the LSTM neural network by introducing the "momentum" regression factor to establish a weighted selective LSTM prediction model based on momentum factor. The model is as follows:

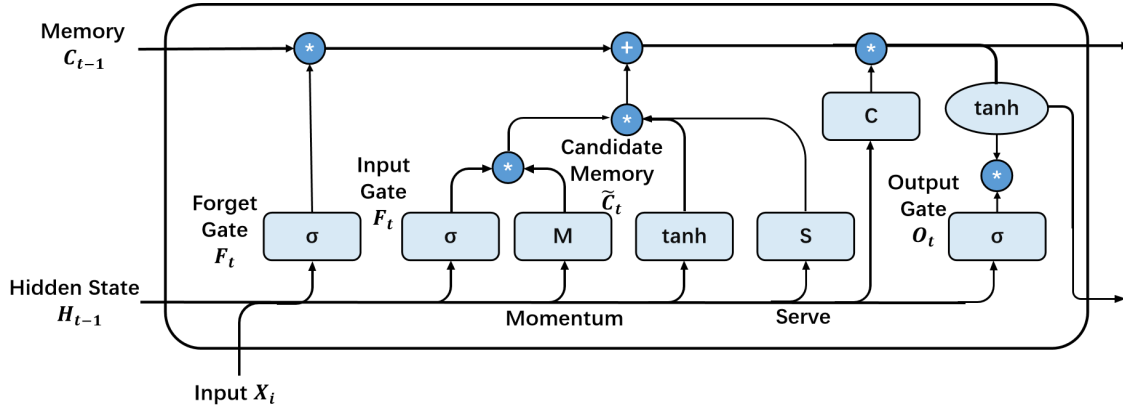


Figure 7: Introduction to the LSTM Model

We improved the LSTM neural network by introducing momentum factor, serving status, and player's competitive level, and constructed corresponding cells. The model formula is as follows:

$$C_t = [F_t * C_{t-1} + I_t * M * S * \tilde{C}_t] * C \quad (5)$$

The training results of the model are presented in the following figure:

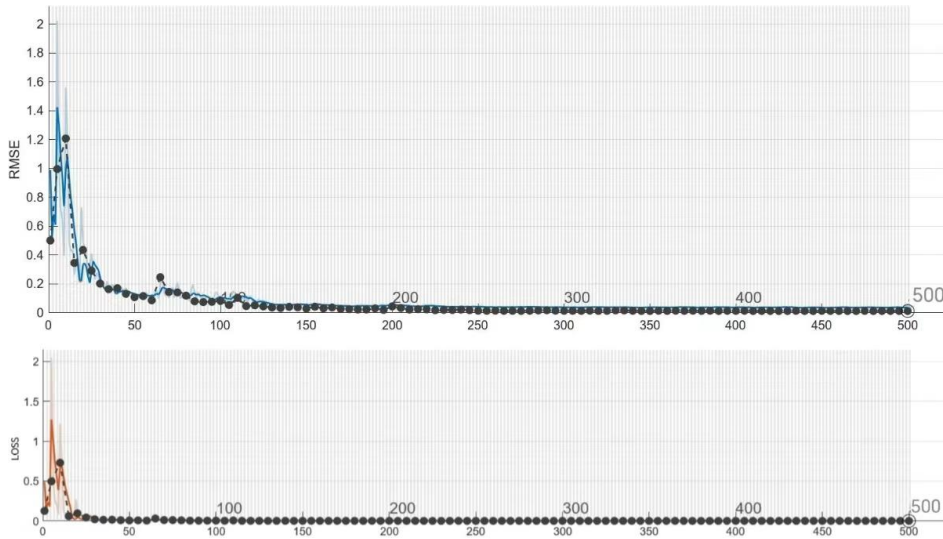


Figure 8: Neural Network Training Results

The model's goodness of fit reached 95.02%, indicating that the model we established has very high accuracy.

Model Fit and Prediction Results:

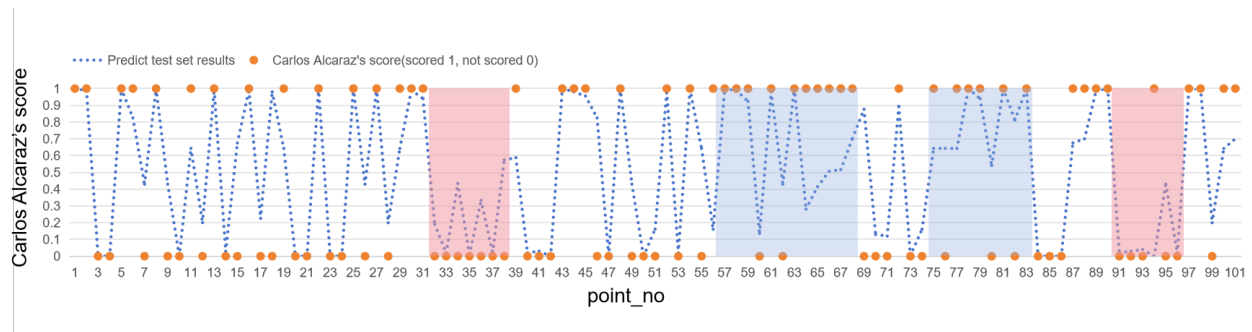


Figure 9: Player Status Prediction Chart

Above is a partial snapshot of the "momentum" and serve composite factor's relationship with player scores. The red segment represents players with a lower momentum level, experiencing significant point losses during this phase. Conversely, the blue segment indicates players with a higher momentum level and a notable improvement in scoring ability.

The data demonstrates that the model accurately identifies the scoring ability of players in their current state. Additionally, the magnitude of scoring based on momentum allows us to assess the players' overall performance.

4 Resolution of Problem II

4.1 Correlation Analysis of Model Indicators

Player scoring is influenced by external factors. We explored the correlation between external factors and the "momentum" factor to assess whether momentum independently affects player scores. The JB test was applied to the data, yielding a p-value of 0.037, suggesting that this variable satisfies a normal distribution.

pearson	sprint	racket	serve width	serve depth	return depth	in net	speed	score
consecutive points	-0.0604	-0.1190	0.0608	-0.0013	-0.0185	-0.0407	-0.0883	0.0815
p_ace	-0.1318	-0.1096	-0.0972	0.1859	0.0723	-0.0501	-0.0076	-0.0459
p_winner	0.0169	0.1178	-0.0816	0.0539	0.0330	0.2802	0.0039	0.8375
ACE	-0.0775	-0.0710	-0.0912	0.1176	0.0295	-0.0597	-0.0322	-0.0516
break_pt_won	-0.0127	-0.0012	-0.0523	0.0452	0.0679	-0.0412	-0.0085	0.0148
behind	-0.0204	-0.0069	0.0550	0.0401	0.0241	0.0346	0.0310	0.0104
double_fault	-0.0661	-0.0540	-0.1160	0.1346	0.1121	0.0727	0.0949	0.0642
unf_err	0.0232	0.0441	-0.0506	0.1070	0.0973	-0.0234	0.0291	-0.1079

Table 3: Indicator Correlation Analysis

From the correlation analysis, it is evident that the eight factors influencing momentum, the two factors affecting physical fitness, and the six factors influencing tactical richness exhibit low correlation. Therefore, it is considered that there is no significant correlation between these factors. This strengthens the assertion that momentum acts as an independent factor influencing player scores.

4.2 Case Study: Impact of Momentum on Match Trends

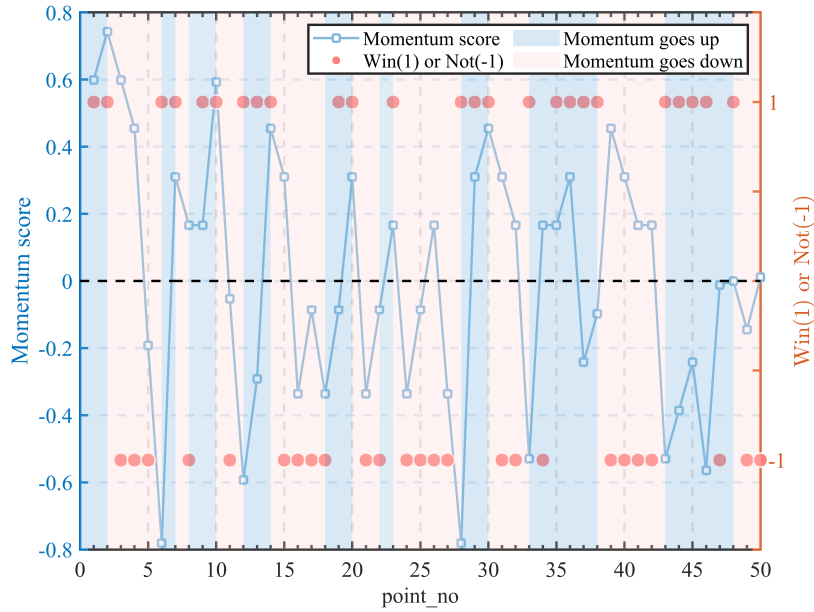


Figure 10: Impact of Momentum on Match Trends

Through the analysis of the above figure, we can clearly observe that when a player's momentum is on the rise, scoring ability significantly improves, and the player can secure the majority of match victories. Conversely, when a player's momentum declines, there is a noticeable increase in points lost. This observation confirms that momentum indeed has a substantial impact on a player's scoring ability, contradicting the coach's viewpoint.

4.3 Monte Carlo Simulation to Validate Model Accuracy[2]

The Monte Carlo model[3] is a stochastic simulation method used to verify results by simulating event occurrences. We randomly generated "momentum" factor data for players through Monte Carlo simulation and then conducted an error test on the objective function.

We set the number of simulations per run to 1000 and performed ten sets of simulations. The average deviation reached 93.21%. In contrast, the goodness of fit for our established weighted selective LSTM neural network[?] prediction model based on the momentum factor reached 95.02%. From a statistical perspective, the impact of momentum on a player's scoring ability is inevitable. Therefore, we consider the coach's viewpoint to be incorrect, as confirmed through Monte Carlo simulation.

5 Resolution of Problem III

5.1 Modeling Factors Affecting Fluctuations

After organizing, we categorized the factors influencing a player's real-time scoring ability into the following five aspects:

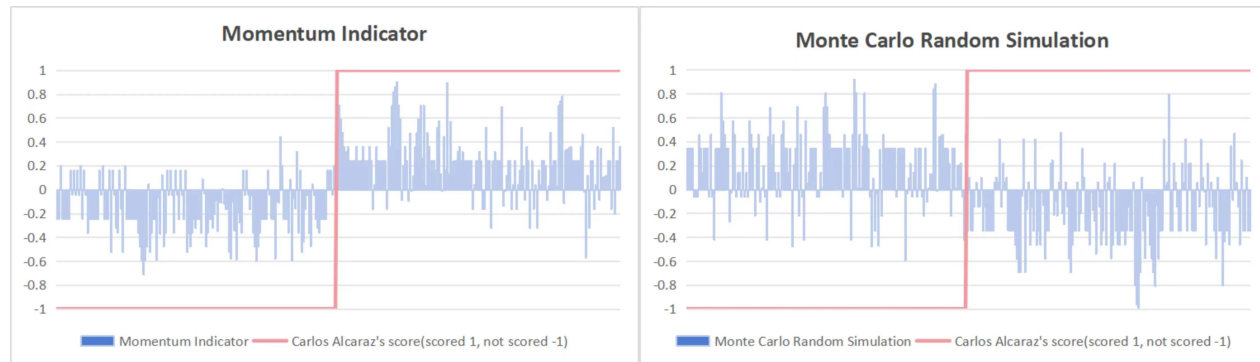


Figure 11: Monte Carlo Simulation Validation

Part 1: Player Skill Level (C)

We analyzed five indicators deter

1. Entropy Weight Method to Determine Indicator Contributions

We uniformly selected match data from four semi-finalists, two quarter-finalists, two round-of-16 players, and two round-of-32 players to ensure the representativeness of our evaluation model. We used the entropy weight method to calculate the information entropy of the indicators, determining weights based on the data's degree of variation.

Step1:Standardization of Evaluation Indicators

Step2:Evaluation Indicator Positive Transformation

For eliminating the influence of different dimensions on the evaluation results, standardization was performed on all data indicators.

Step3:Calculation of Information Entropy for Each Indicator:

Step4:Normalization of Indicator Weights:

We obtained the following weights for each indicator:

X6	X10	X19	X16	X17
0.202233	0.334686	0.158201	0.129507	0.175372

2. TOPSIS Method for Player Skill Level Score

We utilized the TOPSIS method to normalize scores for players, evaluating schemes based on the distance from the ideal optimal solution and the worst solution. This method helps avoid subjectivity in the data assessment.

Step1:Definition of Maximum and Minimum Values:

Step2:Calculation of Euclidean Distance:

Step3: Score normalization

The scores are then normalized, resulting in the player's skill level score.

NAME	Alexander Bublik	Christopher Eubanks	Daniil Medvedev	Alexander Zverev	Frances Tiafoe	Carlos Alcaraz	Jannik Sinner	Novak Djokovic	Andrey Rublev	Nicolas Jarry
Competitive level rating	0.056163	0.092578	0.095063	0.017309	0.020158	0.225145	0.12625	0.233464	0.092902	0.04095

Results indicate that the player's overall score aligns closely with the match outcomes, suggesting high reliability in our evaluation model. Interestingly, in the final, Alcaraz, who defeated Djokovic, had a lower score than his opponent. This anomaly highlights the significant roles played by a player's momentum and psychological resilience in influencing match outcomes. It indirectly reflects the validity of our research.

Part 2: Tactical Richness Evaluation Model (T)

We converted factors such as the player's consecutive five serves into their degree of variation to measure tactical changes during the match. A multiple linear regression model was established for analysis. Tactical changes were categorized into serving and receiving serves, and the impact factors for each were analyzed and processed.

$$t_r = \begin{cases} t_1 : \text{Whether the direction of the serve changes} \\ t_2 : \text{Whether the depth of the serve changes} \\ t_3 : \text{Whether the return depth has changed} \\ t_4 : \text{Whether the player made it to the net} \\ t_5 : \text{Whether the ball speed changes} \\ t_6 : \text{The reason of one score} \end{cases} \quad (6)$$

$$tr = t_1 f(x_1) + t_2 f(x_2) + \dots + t_6 f(x_6) = \sum_{i=1}^6 t_i f(x_i) \quad (7)$$

Despite the clear correlation between tactical richness and a player's scoring ability, its contribution to whether a player scores is relatively low. To avoid losing relevant data, we utilized the entropy weighting method to specifically describe the player's tactical richness indicators.

For serving rounds, the model is:

$$F_1(x) = 0.25914 * x_1 + 0.28558 * x_2 + 0.04755 * x_3 + 0.14799 * x_4 + 0.25974 * x_5 \quad (8)$$

For receiving rounds, the model is:

$$F_2(x) = 0.12318 * x_1 + 0.29387 * x_2 + 0.58295 * x_3 \quad (9)$$

Part 3: Physical Fitness Assessment Model (Ps)

We collected information on a player's sprint distance and racket swings during the match, converted it into the average level within consecutive five points, and analyzed it. Based on regression analysis results, we established the player's physical fitness status model as follows:

$$Ps = 0.587 - 0.01 * p_1 + 0.025 * p_2 \quad (10)$$

Part 4: Player Momentum Evaluation Capture Model (M)

Based on the analysis from the first question, we established the player's momentum evaluation capture model as follows:

Part 5: Player Serving (S)

We define whether serving as:

$$S = \begin{cases} 0, \text{not serve} \\ 1, \text{serve} \end{cases} \quad (11)$$

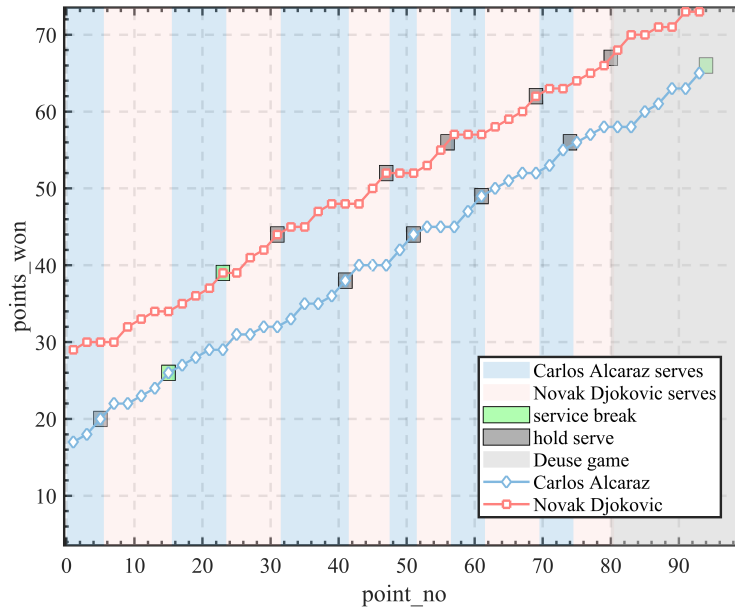
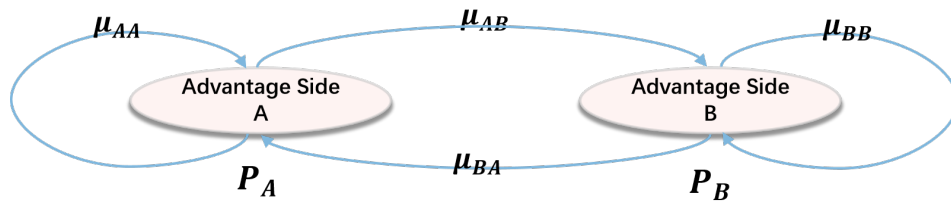


Figure 12: Player Serving Advantage Chart

5.2 Prediction of Fluctuations Based on Markov Chain and Multivariate Nonlinear Regression

Step1: Markov Chain for Determining Inherent Advantage Probability

A Markov chain is a random process that undergoes transitions from one state to another in a state space, and after a certain number of transitions, a stable probability distribution independent of the initial state distribution can be obtained. By referencing player ratings on the ATP official website, we normalized the data of Alcaraz and Djokovic from the final match to form their transition probability matrix $[0.5127 \ 0.4873, 0.4717 \ 0.5283]$ after normalization.



We calculated the following data as the initial state distribution of player ability values: $[0.5577, 0.4423]$. Setting the number of iterations to 3000, we computed that the steady-state probability values for the advantage of both players are $[0.46753, 0.53247]$. We use this probability distribution as the inherent advantage probability distribution for the match, denoted as π .

Step2: Multivariate Nonlinear Regression Analysis for Fluctuation Factors[?]

Due to the complex relationships among fluctuation factors, we separated them and constructed nonlinear regression equations.

Tactical Richness (T):

Exhibits strong significance in influencing score flow, requiring no further processing.

Momentum (M):Constructed the function $f_1(m) = \exp(m)$.

Endurance (Ps):Constructed the function:

$$f_2(p) = 2 + \sin(e^{15p} * 0.05 - 5) \quad (12)$$

Through regression of the above nonlinear regression equations, we obtained the final regression equation:

Our prediction results are as follows:

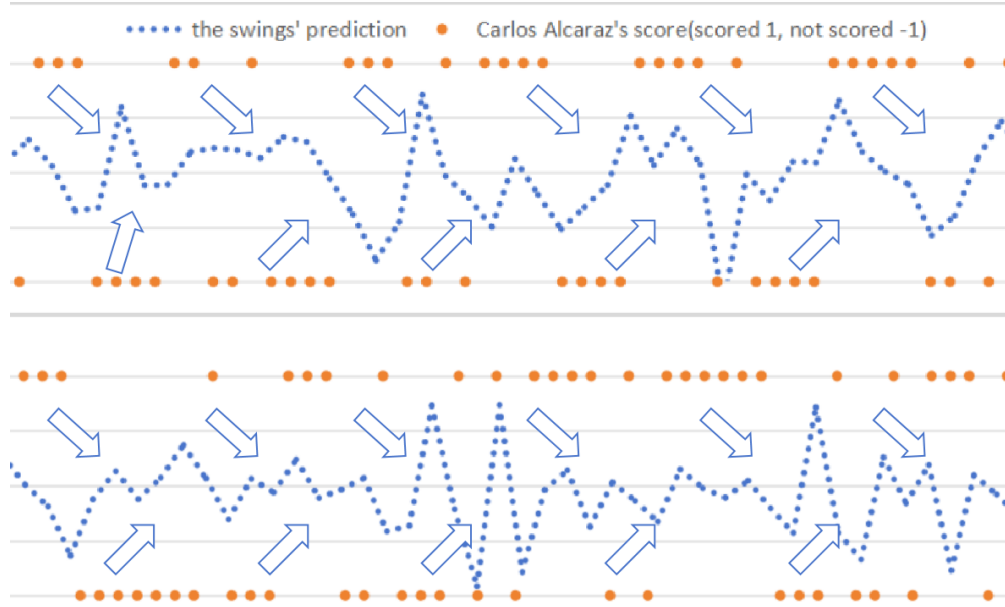


Figure 13: Verification of the Fluctuation Prediction Model

By examining the predicted results, we can observe that our model accurately forecasts the fluctuations in the game. Hence, we conclude that our model possesses a high level of accuracy.

5.3 On-court Strategies Based on Volatility Analysis Recommendations for Players

Due to the significant correlation between momentum and scoring, recommendations are provided for our team players to address potential disadvantages in different momentum situations during a match. These suggestions aim to better utilize the impact of momentum and gain control over the match situation.

When the opponent serves:

- Adopt a conservative strategy.
- Flexibly use tactics to seize points during the opponent's service game.
- Preserve energy and, upon entering our service game, shift suddenly to an offensive strategy to suppress the opponent.

During crucial points or tiebreakers:

- If our player struggles under pressure, adopt an aggressive approach to disrupt the opponent's rhythm during key points.
- Utilize quick serves during tiebreakers to create pressure when serving.

When the opponent aggressively attacks:

- Employ strategies such as lobs to delay the match and slow down the pace.
- Exploit opponent weaknesses, such as attacking their backhand, to consume their energy.

When the opponent has a substantial lead:

- Evaluate personal stamina.
- Strategically concede a few points to conserve energy for a potential comeback.
- In the final set or when the match outcome is critical, consider an aggressive approach to control the situation and seek opportunities for a comeback.

6 Resolution of Problem IV

6.1 Model Testing

We selected the matches in the Round of 16 for data analysis, and the analysis results for some matches are shown in the figure below:

In the above figure, the scatter points of the model are concentrated around (0,0) and (1,1), indicating that our match fluctuation function can predict the true values of scores with a high probability. Through ROC analysis to optimize and find the best threshold, classifying the fluctuation function values, the classification results show an accuracy of over 85%. We believe that our model has high accuracy.

6.2 Expansion of Model Factors

- **Court Adaptability:** Different court surfaces have varying effects on ball bounce and speed. Players' adaptability to different courts also influences their scoring ability, factors like wind speed, temperature, and humidity will ultimately influencing match results.
- **Player Health:** The physical health of players, including injuries and fatigue levels, directly affects their performance in matches.
- **Home and Away Factors:** Playing at home may provide a psychological advantage, while playing away may pose challenges such as dealing with audience emotions and adapting to different environments.

The impact of these factors should not be ignored, however, comprehensive data could not be fully collected. We hope to enhance our model by introducing the mentioned data, making the analysis more comprehensive and accurate.

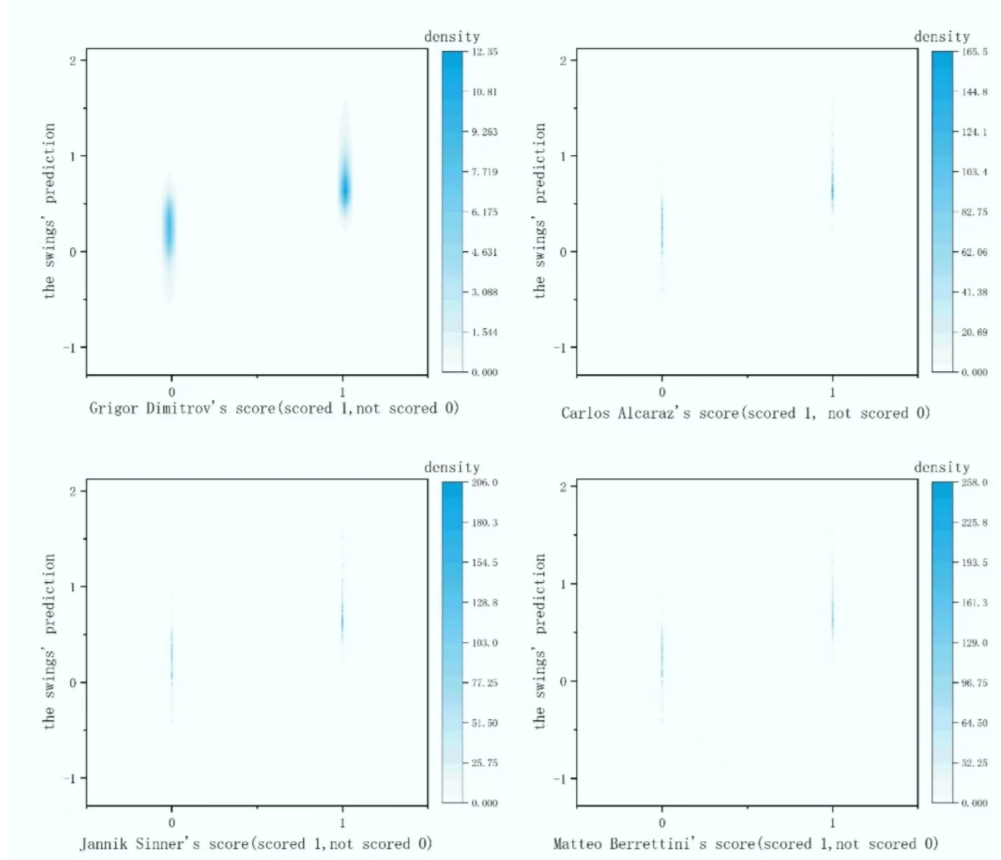


Figure 14: Model test results

6.3 Model Generalization

Part 1: Generalization of the Model in Table Tennis Matches

In table tennis matches, the sprint distance is short, and the duration is brief, with lower demands on endurance compared to tennis matches. However, the faster pace and more diverse scoring techniques lead to a greater emphasis on the richness of tactics and the player's own abilities. When generalizing the model, these factors are taken into consideration.

Improvement of the Table Tennis Match Model : Analyzing the data of the 2023 World Table Tennis Championships final match, we improved the model for predicting the scoring dynamics in table tennis matches. Other parts of the model were retained while the non-linear equation for physical fitness was reconstructed as follows:

$$f(x) = 0.8 - \cosh(e^x * 0.02 + 0.5) \quad (13)$$

As tactical richness is influenced by various scoring techniques, this generalized model introduces factors such as backhand topspin ratio, topspin with sidespin ratio, and loop, constructing the equation as follows:

$$f(x) = 0.026x_2 + 3.4x - 0.091 \quad (14)$$

After processing the nonlinearized data and the distribution of inherent advantages, calculations were performed, and the results were verified, indicating that the model's generalization to table tennis matches is relatively accurate.

Part 2: Model Generalization in Tournaments

Model Adjustments: Taking the example of the Dallas Open in ATP250, we analyzed the match data between Emilio Nava and Aidan Mayo in the 2024 Men's Singles 1st Round Qualifying. Topsis entropy weighting method was used to score various factors. Notably, the weights of several factors, such as first serve, unforced errors, double faults, and net points won, changed significantly. The model achieved high accuracy.

Analysis of Weight Changes Comparing the weight impact in Wimbledon with that in a tournament, double faults and first serve had increased weights in tournaments. This explains the weaker pressure resistance of players and the greater impact of momentum in tournaments. However, unforced errors and net points won had decreased weights, reflecting lower player abilities and tactical richness in tournaments. Therefore, when introducing the model into tournaments, it is necessary to reevaluate and adjust the weights to enhance model accuracy.

Part 3: Model Generalization in Women's Matches

We tested the model using the match data between B. Pera and A. Krueger in the WTA Mubadala Abu Dhabi Open, finding a reduced prediction accuracy of 72.383%, significantly different from the 85% or higher accuracy in men's matches. Thus, a detailed analysis of the differences between men's and women's tennis and the adjustment of weights are essential for successfully generalizing the model. Statistical analysis of WTA women's singles matches reveals fewer points decided by high-strength techniques, such as net approaches and baseline rallies, in women's matches. Instead, female players often influence the match through varied tactics and clever strategies. These factors need to be considered when extending the model to women's matches, and corresponding adjustments or reweighting of influencing factors are required to ensure the accuracy of the model's analysis in real situations.

Part 4: Generalization of the Model to Different Court Surfaces[1]

Tennis courts are mainly categorized as clay, hard, and grass courts, each having a significant impact on player performance. The Wimbledon Championship in the dataset is played on grass courts, and the prediction accuracy drops when applied to other court surfaces. Therefore, it is crucial to consider the influence of different court surfaces on the model.

The impact of court surfaces on ball speed is a key difference among different courts. Grass courts, as involved in the model, are the fastest with minimal speed decay. Clay courts have loose surfaces, resulting in slower ball speed and greater speed decay. Hard courts have a more regular bounce, requiring higher serving quality. An analysis of the typical US Open on hard courts reveals a 24.3% increase in the impact of ACE serves on scoring factors compared to grass court data. Clay courts slow down ball speed and produce higher bounce, leading to longer baseline rallies and greater challenges for players in terms of baseline techniques and endurance. Data from the French Open also supports these observations.

Considering these differences, we aim to obtain test data for different court surfaces and establish a court surface factor to improve the model's generalization.

7 Sensitivity Analysis

Conducting sensitivity analysis on the fluctuation function involves examining variables that impact the model, including the inherent advantage probability distribution of players and the fluctuation function. As the initial state distribution in the Markov chain does not affect the final state distribution, sensitivity analysis is performed on the variable with the maximum uncertainty in the multivariate nonlinear regression, namely, the fitness value, with a coefficient of 0.08 as the original coefficient.

To assess the stability of the model, the coefficient for the richness of tactics in the multivariate nonlinear regression is varied around the original coefficient. This is achieved by observing the standard errors and

P-values in the regression results for different coefficients.

The magnitude of the change relative to the original coefficient	Standard Error	P
-10%	0.094	0.019*
-5%	0.038	0.018**
0%	0.021	0.020**
5%	0.075	0.005*
10%	0.14	0.073*

Within the range of plus or minus 10%, the standard error increases with the growing fluctuations, and its significance is somewhat affected. However, the standard error still remains within a relatively small range. Through significance analysis, it is evident that the richness of tactics continues to exhibit significance in the regression results. Therefore, it can be concluded that the model we have constructed is stable.

8 Strengths and weaknesses

8.1 Strengths

- The quantification of player data through data clustering and factor analysis, leading to the development of an athlete's competitive level evaluation model, is commendable.
- The improvement of the LSTM neural network by incorporating momentum factors, serving status, and athlete competitive levels into corresponding cells significantly enhances the model's accuracy.
- The persuasive nature of the results is reinforced by employing Monte Carlo simulation to analyze the factors influencing momentum.
- The establishment of objective indicators for model evaluation, extracting relationships between intrinsic factors within the indicators, contributes to the objectivity and persuasiveness of the model.
- The utilization of fluctuation prediction based on Markov chains and multivariate nonlinear regression demonstrates a high level of model accuracy.

8.2 Weaknesses

- Due to the limited nature of the provided data and the focus on professional players, the model may experience a noticeable decrease in accuracy when predicting outcomes for youth matches and considering training data, as it does not capture information about young players.
- The constraints of time prevented the exploration of more relationships between indicators, leading to the model's inability to identify additional underlying connections among the data.
- The assumption in the paper that other factors remain constant doesn't align with the actual conditions of tennis matches, which are influenced by variables such as weather and court conditions. The model has not incorporated these variables into its analysis.

Memorandum

To: Coaches

From: Team 2429459

Date: February 5, 2024

Subject: Exploration of Momentum in Tennis and Model Recommendation

Dear Coaches,

Greetings! We are a group of university students passionate about tennis. After watching the 2023 Wimbledon Gentlemen's final, our interest in the impact of momentum on the game was ignited. Through exploring interesting phenomena in the matches and drawing conclusions, we wish to share our findings with you. We hope that our research can assist players in better understanding and utilizing momentum for a favorable impact on their matches. This includes timely adjustments to game tactics, effectively utilizing and changing momentum for improved results. Additionally, our insights aim to help in crafting more suitable training plans to enhance players' skills. Below are our key findings.

Firstly, we observed an intriguing phenomenon. In the final, the number of aces and the scoring situation of the players showed a negative correlation. This finding contradicts the common sense belief that hitting aces motivates players. We suggest that high-quality serves may lead to high-quality returns from opponents, posing a risk of diminishing the server's advantage. Simultaneously, opponents might exhibit better performance in the subsequent points to thwart the server's momentum. Therefore, aces may have a potentially negative impact on the momentum of high-level players.

Extensive data analysis revealed that momentum factors often play a significant role. We improved the LSTM neural network prediction model by incorporating player momentum and competitive status for training, achieving excellent results. We recommend this model, as it can analyze a player's advantage in a match based on their performance. This enhances a player's real-time scoring ability by evaluating the changing states of both players, allowing athletes to deploy optimal tactics at the right moments.

Subsequently, we extracted and analyzed the intrinsic factors of indicators. Through various methods, we established five indicators for a comprehensive model of influencing factors. We then developed a volatility prediction model based on Markov chains and multiple nonlinear regression, proposing strategies for athletes in different scenarios. Our suggestions include:

Recommendation 1: When the opponent's momentum consistently surpasses our player's, adopt a defensive strategy to minimize errors. Employ diverse tactical variations to disrupt the opponent's rhythm and weaken their state by extending the number of rounds.

Recommendation 2: When the opponent serves, consider taking a more aggressive approach on their third or fourth serve. In other points, adopt a conservative strategy, utilizing varied tactics to wear down the opponent's state.

Recommendation 3: After hitting an ace, swiftly heighten attention and change serving tactics to prevent consecutive point losses.

Recommendation 4: Through targeted training, identify conditions and timings for capturing player momentum. Adjust tactics to enable players to enter a higher momentum state during crucial sets, enhancing their ability to score on important points.

We hope our research findings contribute to the understanding and application of momentum in tennis. Thank you for considering our insights, and we look forward to any feedback or further discussion.

Sincerely,

Team 2429459

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